



UNIVERSITY OF JAFFNA

FACULTY OF ENGINEERING

Department of Electrical & Electronic Engineering

EC8010: Design Proficiency 2026

DESIGN TASK 03 - Preliminary Lab Session

Rotary Inverted Pendulum

You are required to design a controller for “Furuta Pendulum” as shown in the Figure 01 below to balance a rotary inverted pendulum when external disturbances applied. The goal of the control is to bring the pendulum close to the **upright and downright position** regardless of uncertainties or disturbances.

The pendulum and controller stability will be evaluated by applying external disturbances to the vertical rod and its response. To achieve fast response and system stability, careful selection of the motor, controller, driver, sensor, and physical construction of the pendulum **must be considered**.

Constraints: You are allowed to use only one motor and one feedback sensor. You are only allowed to apply controller and stability criteria that you have studied in the past semester.

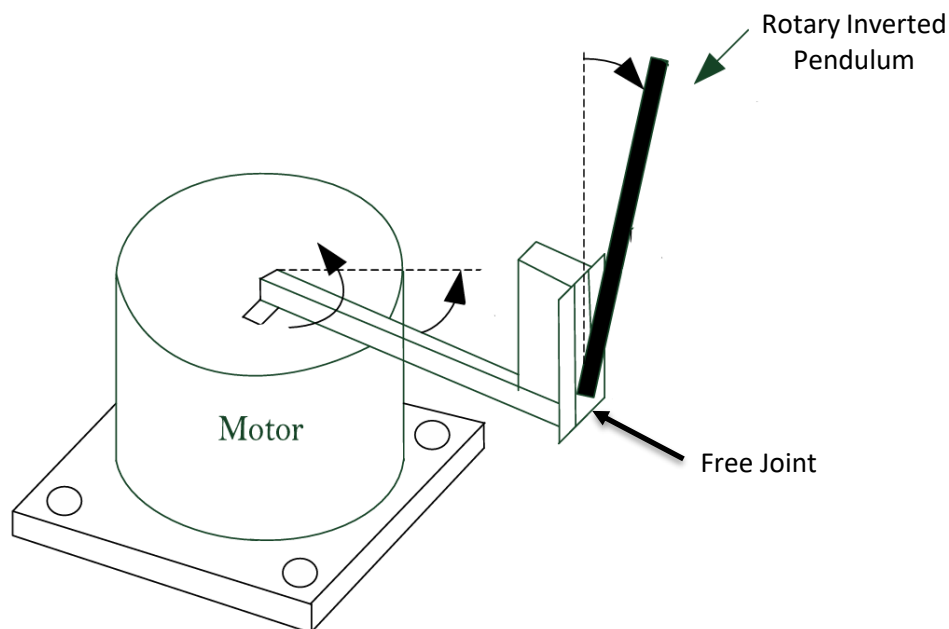


Figure 01